

Phase-3 Submission Template

Student Name: santhosh.s

Register Number: 731323104043

Institution: JKKN college of engineering and technology

Department: B.E CSE

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Github Repository

Link: <https://github.com/santhoshsarth/batch03>

1. Problem Statement

Stock market prediction is highly challenging due to its dynamic, non-linear, and volatile nature. Traditional methods often fail to capture complex patterns in historical data. This project aims to leverage AI-driven time series analysis to accurately forecast stock prices, enabling smarter investment decisions.

2. Abstract

This project explores AI-driven time series analysis to predict stock prices with greater accuracy. By leveraging machine learning models on historical market data, it uncovers hidden trends and patterns. The approach aims to enhance decision-making in stock trading through reliable and data-informed forecasts.

3. System Requirements

Hardware

- ✓ Hardware: Minimum 8GB RAM, i5 processor or higher, and at least 100GB of storage.
- ✓ Operating System: Windows 10/11, Linux (Ubuntu 20.04+), or macOS.

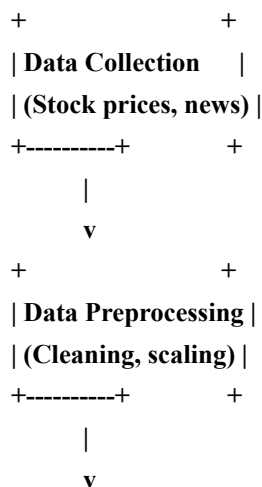
Software:

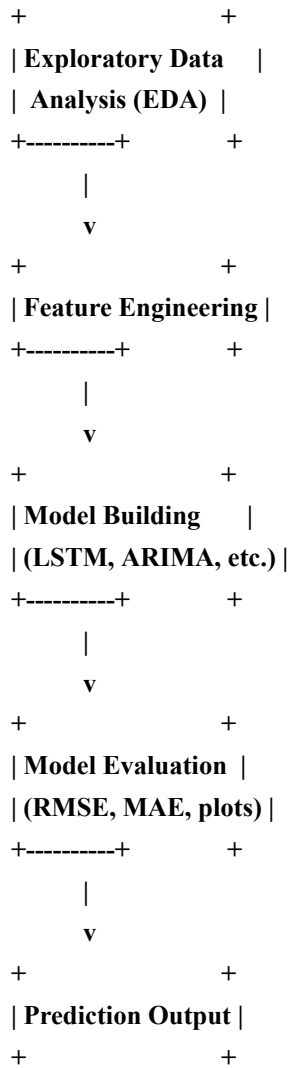
- ✓ Software: Python 3.8+, Jupyter Notebook/Anaconda, and libraries like NumPy, Pandas, Scikit-learn, TensorFlow/Keras.
- ✓ Internet Connection: Required for accessing stock data APIs and installing dependencies.

4. Objectives

1. To develop an AI-driven model for accurate stock price prediction using time series analysis.
2. To analyze historical stock market data and identify meaningful trends and patterns.
3. To compare the performance of different machine learning algorithms for forecasting.
4. To assist investors in making informed decisions through data-driven insights.

5. Flowchart of Project Workflow





6. Dataset Description

- ✓ **Source:** APIs like Yahoo Finance or Alpha Vantage.
- ✓ **Features:** Date, Open, High, Low, Close, Volume, and Adjusted Close.
- ✓ **Time Range:** series analysis and model training

7.Data Preprocessing

- ✓ Checked and confirmed: No missing values
- ✓ Normalized pixel values
- ✓ Reshaped data

8. Exploratory Data Analysis (EDA)

- Visualize historical stock prices using line plots
- Use correlation heatmaps to analyze
- Examine distribution plots and boxplots
- Apply autocorrelation and partial

9.Feature Engineering

- 1.Create technical indicators like RSI, MACD, and moving averages.
2. Generate lag features and rolling statistics to capture temporal patterns.
3. Add time-based and external features such as day, month, and sentiment scores.

10 .Model Building

1. Select suitable models like ARIMA, LSTM, or XGBoost for time series forecasting.
2. Train models using processed data while preserving time order.
3. Tune hyperparameters to optimize predictive performance.

11.Model Evaluation:

1. Use metrics like RMSE, MAE, and MAPE to assess accuracy.
2. Validate models using time-based cross-validation.
3. Compare model performance on test data to select the best approach.

12. Deployment

Platform: Streamlit Cloud

Include:

- ✓ Public link
- ✓ Screenshot of UI
- ✓ Sample input/output (user-drawn digit → predicted label)

13. Source code

```
import numpy as np

import pandas as pd

import matplotlib.pyplot as plt
from sklearn.preprocessing import MinMaxScaler
from keras.models import Sequential
from keras.layers import LSTM, Dense
from sklearn.metrics import mean_squared_error
import yfinance as yf

# 1. Load Stock Data
data = yf.download('AAPL', start='2015-01-01', end='2024-12-31')
close_prices = data['Close'].values.reshape(-1, 1)

# 2. Normalize Data
scaler = MinMaxScaler()
scaled_data = scaler.fit_transform(close_prices)

# 3. Create Time Series Dataset
def create_dataset(dataset, time_step=60):
    X, y = [], []
    for i in range(len(dataset)-time_step-1):
        X.append(dataset[i:(i+time_step), 0])
        y.append(dataset[i+time_step, 0])
    return np.array(X), np.array(y)

X, y = create_dataset(scaled_data, 60)
X = X.reshape(X.shape[0], X.shape[1], 1)

# 4. Split Data
split = int(len(X) * 0.8)
X_train, X_test = X[:split], X[split:]
y_train, y_test = y[:split], y[split:]

# 5. Build LSTM Model
model = Sequential()
model.add(LSTM(50, return_sequences=False, input_shape=(60, 1)))
model.add(Dense(1))
model.compile(optimizer='adam', loss='mean_squared_error')

# 6. Train Model
model.fit(X_train, y_train, epochs=10, batch_size=32, verbose=1)

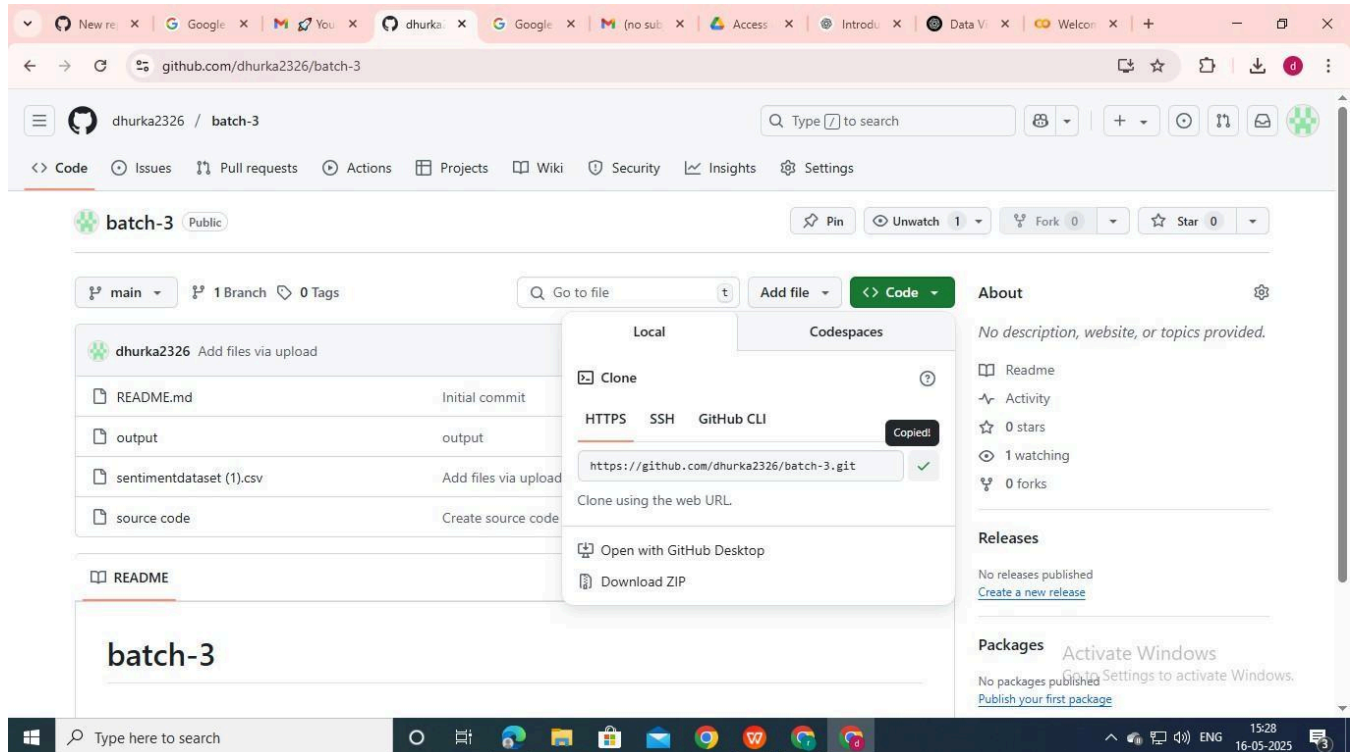
# 7. Predict and Evaluate
y_pred = model.predict(X_test)
y_pred = scaler.inverse_transform(y_pred.reshape(-1, 1))
y_test_actual = scaler.inverse_transform(y_test.reshape(-1, 1))

# 8. Plot Results
plt.figure(figsize=(10,6))
plt.plot(y_test_actual, label='Actual Price')
```

```
plt.plot(y_pred, label='Predicted Price')  
plt.title('Stock Price Prediction')  
plt.xlabel('Time')  
plt.ylabel('Price')  
plt.legend()  
plt.show()
```

9. Calculate RMSE

```
rmse = np.sqrt(mean_squared_error(y_test_actual, y_pred))  
print("RMSE:", rmse)
```



14. Future scope

- AI-driven stock prediction can be enhanced with real-time data integration
- inclusion of sentiment analysis from social media and news, and the use of more advanced
- deep learning architectures like Transformers for improved forecasting accuracy.

15. Team Members and Role

Member 1 – Dhurka

Devi.D Role: Model

Developer

Responsibilities:

- ✓ Tune hyperparameters for improved accuracy
- ✓ Evaluate model using accuracy, F1-score, confusion matrix
- ✓ Integrate and test final project flow

G.N Madhumita– Data Specialist

Role: Data Preprocessing & EDA

Responsibilities:

- ✓ Normalize and reshape image data
- ✓ Perform EDA with visualizations (histograms, heatmaps)
- ✓ Document insights from data analysis

Chandru.S– Deployment Engineer

Role: UI & Deployment

Responsibilities:

- ✓ Link trained model to the frontend for predictions
- ✓ Deploy on cloud platform (Streamlit Cloud or Hugging Face Spaces)
- ✓ Capture UI screenshots and provide deployment link

Santhosh.S– Documentation Expert

Role: Report & Presentation

Responsibilities:

- ✓ Draft report sections like problem statement, abstract, objectives, etc.
- ✓ Create the flowchart and future scope
- ✓ Prepare submission-ready presentation

